VISVESVARAYA TECHNOLOGICAL UNIVERSITY

JNANASANGAMA, BELAGAVI-590018

An Project Report on

"Solar Powered Wireless Power Transfer System for Electric Ships"

Submitted in partial fulfillment for the award of degree of

Bachelor of Engineering

In

Department of Electrical and Electronics Engineering

for the **Academic Year: 2025-2026**

Submitted by

Deepti	**(1NT22EE024)**
Harshitha D R	**(1NT22EE034)**
Meghana B S	**(1NT22EE053)**
Monika N R	**(1NT22EE057)**
N Srividya	**(1NT22EE060)**

Under the Guidance of

Dr . Rajarajan Ramasamy
Assistant Professor
Dept. of Electrical and Electronics Engineering

NITTE EDUCATION TRUST
NITTE MEENAKSHI INSTITUTE OF TECHNOLOGY

Yelahanka, Bengaluru-560064, Karnataka, India

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Certificate

This is to certify that the project work entitled *"*Solar Powered Wireless Power Transfer System for Electric Ships*"* has been carried out by Deepti(1NT22EE024), Harshitha.D.R (1NT22EE034),Meghana B.S(1NT22EE053), Monika N.R(1NT22EE057) and N.Srividya (1NT22EE060)**,** bonafide students of ***Nitte Meenakshi Institute of Technology****,* in partial fulfillment of the requirements for the award of the degree of **Bachelor of Engineering** in the **Department of Electrical and Electronics Engineering**, under **Visvesvaraya Technological University**, Belagavi, during the academic year **2025– 2026.**

The project report has been examined and approved as it meets the academic requirements specified under the autonomous scheme of **Nitte Meenakshi Institute of Technology** for the said degree.

Signature of the Guide	Signature of the HoD	Signature of the Principal
Dr. Rajarajan Ramasamy Assistant Professor Nitte Meenakshi Institute of Technology, Bengauru-560064	**Dr. Samanvita. N** Professor & HoD Nitte Meenakshi Institute of Technology, Bengauru-560064	**Dr. H C. Nagaraj** Principal Nitte Meenakshi Institute of Technology, Bengauru-560064

External Viva-Voce

Name of Examiners	***Signature with Date***
1. ……………………………..	**…………………………..**
2. ……………………………..	**…………………………..**

Acknowledgement

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Date- 21/05/2026
Place - Bangalore

Abstract

This project presents the design and development of a solar-powered wireless power transfer system intended for charging electric ships in marine environments. The system provides a contactless charging solution that eliminates the safety risks and maintenance issues associated with conventional wired charging in water environments.

The proposed system integrates floating solar photovoltaic panels with a Maximum Power Point Tracking controller to efficiently harvest solar energy. This energy is stored in a battery and then converted to high-frequency alternating current, which is transmitted wirelessly through resonant inductive coupling between a transmitter coil at the charging station and a receiver coil mounted on the ship. The received power is rectified, filtered, and used to charge the ship's battery.

An ESP32 microcontroller-based monitoring system continuously measures voltage, current, and power using sensors and displays these values on a 16×2 LCD interface. Additional solar panels installed on the ship provide supplementary power during daytime operation, improving overall system reliability.

Experimental results demonstrate that power transfer efficiency depends significantly on coil separation distance, alignment accuracy, and operating frequency. The system successfully validates the feasibility of combining solar energy with wireless charging technology for marine applications, offering a cleaner and safer alternative to conventional charging methods

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Chapter 1: Introduction

1.1 Background

The global transition toward sustainable energy has accelerated the development of electric vehicles across all transportation sectors, including maritime vessels. According to the International Maritime Organization (IMO), the shipping industry accounts for approximately 2.89% of global greenhouse gas emissions, prompting urgent calls for decarbonization [1]. Electric ships offer a promising solution for reducing harmful emissions and decreasing dependence on fossil fuels in marine transportation.

Wireless Power Transfer (WPT) technology has emerged as a transformative solution for electric vehicle charging, eliminating the need for physical connectors that are prone to wear, corrosion, and safety hazards. WPT systems based on resonant inductive coupling can transfer energy efficiently across air gaps, making them particularly suitable for marine environments where traditional plug-in charging is problematic [2].

1.2 Motivation

The motivation for this project stems from several critical observations:

1. Limitations of Wired Charging in Marine Environments

Conventional wired charging methods present multiple challenges when deployed in marine settings. Electrical cables and connectors are continuously exposed to water and salt, leading to rapid corrosion of metallic contacts. This degradation not only increases maintenance costs but also creates serious electrical safety hazards, particularly during wet docking conditions. Physical connections are difficult to manage near moving vessels, and the constant wear from repeated plug-in operations reduces system reliability [3].

2. Potential of Wireless Power Transfer

WPT technology offers distinct advantages over wired alternatives. By eliminating physical contact points, WPT systems are inherently more robust against corrosion and water ingress. The absence of cables simplifies docking procedures and enables automated charging operations. Furthermore, resonant inductive coupling can achieve high efficiency at practical distances, making it feasible for marine applications where vessel positioning may vary [4].

3. Integration with Renewable Energy

The combination of WPT with solar energy creates a self-sufficient charging infrastructure that can operate independently of the conventional power grid. Floating solar photovoltaic panels can be deployed on water surfaces, utilizing otherwise unused space while providing clean energy for charging stations. This integration supports off-grid operations in remote locations where traditional charging infrastructure is unavailable or prohibitively expensive to install [5].

1.3 Scope of the Project

This project encompasses the following key areas:

1. Design of Resonant Inductive WPT System: Development of a resonance-based wireless power transfer system operating at optimized frequency for marine applications.
2. Solar Energy Integration: Implementation of floating solar photovoltaic panels with MPPT charge controller for maximum energy harvesting.
3. Monitoring and Control: Development of an ESP32-based real-time monitoring system for voltage, current, and power measurement using I2C LCD display.
4. Performance Analysis: Experimental evaluation of system efficiency under varying conditions including coil separation distance, misalignment angles, and load variations.
5. Backup Power Integration: Demonstration of hydrogen fuel cell concept (using magnesium and graphite electrodes) as an emergency power source.

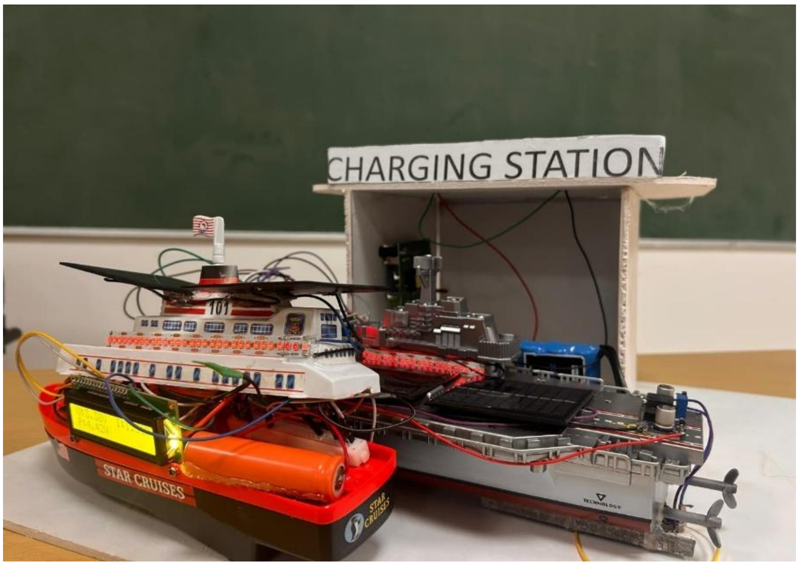

Fig 1.3.1 Solar Powered wireless power transfer system for electric sh

Chapter 2: Literature Survey

2.1.1 Design and Implementation of Resonance Based Wireless Power Transfer System

Authors: Koustubh Jadhav, Vishal Urlam, Rahul Harer, Shivani Naik, Dr. Anjali Deshpande, and Hemant Jadhav.

Abstract:

This paper explains the design and development of a resonance-based wireless power transfer system. The main aim of the project is to transfer electrical power without using physical wires. The system operates using high-frequency resonant inductive coupling, where power is transferred between transmitter and receiver coils through an air gap. A series-series compensation method is used to improve efficiency and power transfer capability. The developed prototype works at a switching frequency of 26 kHz and delivers nearly 15 W of power. Experimental analysis shows that the system can transfer power effectively up to an air gap of 8 cm with a maximum efficiency of about 68%. The proposed system can be useful in applications such as wireless charging of electric vehicles and portable electronic devices.

Outcome / Conclusion:

The wireless power transfer system was successfully designed and tested. The developed prototype achieved efficient power transfer without direct electrical contact between the coils. The system delivered a maximum output power of around 10 W with an efficiency close to 68%. It was observed that the efficiency decreases as the distance between the coils increases. The project demonstrates that resonance-based wireless charging can be used for future charging applications such as electric vehicles and consumer electronics. Future improvements like using Litz wire, SiC MOSFETs, and higher switching frequencies can further enhance the system performance.

2.2 5-kW, 96.5% Efficiency Capacitive Power Transfer System With a Five-Plate Coupler: Design and Optimization

Authors: Enguo Rong, Pan Sun, Gang Yang, Jinglin Xia, Zhe Liu, and Siqi Li

Abstract:

Wireless power transfer technology is becoming important in modern applications such as electric vehicles, electric ships, drones, and industrial systems. This paper presents a newly designed five-plate capacitive power transfer (CPT) system that improves power transmission efficiency while reducing system complexity and cost. The proposed design uses single-sided shielding to reduce electric field leakage and improve system performance. In addition, an optimization technique is introduced to increase the efficiency of asymmetric coupler structures. A prototype of the system with a power capacity of 5 kW and a transfer distance of 100 mm was developed and experimentally tested. The results showed that the system achieved a high efficiency of 96.5%, demonstrating reliable and efficient wireless power transfer. The proposed design shows strong potential for practical high-power applications.

Outcome:

The proposed five-plate capacitive power transfer system achieved successful high-power wireless energy transmission with improved efficiency. The developed prototype delivered 5 kW output power with 96.5% efficiency at a transfer distance of 100 mm. The system also showed stable performance, lower power losses, and good tolerance against alignment variations. The results prove that the proposed design can improve wireless charging systems and can be effectively used in future applications such as electric ships, electric vehicles, and industrial wireless power systems.

2.3 Wireless Power Transfer via Strongly Coupled Magnetic Resonances

Authors: A. Kurs, A. Karalis, R. Moffatt, J. D. Joannopoulos, P. Fisher, and M. Soljačić

Abstract:

Wireless power transfer has gained attention as an alternative method for supplying energy without physical connections. This paper presents a technique based on strongly coupled magnetic resonances to transfer electrical energy over a certain distance efficiently. The proposed method uses resonant magnetic fields between two objects to enable energy transfer while minimizing energy loss to the surrounding environment. Experimental analysis demonstrated that power could be transmitted successfully without direct contact and with acceptable efficiency levels. The study highlights the possibility of using this technology in different applications, including consumer electronics, medical devices, and wireless charging systems.

Outcome:

The study successfully demonstrated that electrical energy can be transferred wirelessly using strongly coupled magnetic resonance. The results showed that power could be delivered across a moderate distance while maintaining efficient energy transfer. The research proved the practicality of contactless power transmission and became a significant step toward the development of modern wireless charging technologies. The findings also opened opportunities for future research in wireless energy systems for various industrial and commercial applications.

Chapter 3: Problem Statement

3.1 Problem Statement

The maritime industry is moving toward electric propulsion to reduce pollution. However, electric boats and ships cannot be widely adopted until reliable charging infrastructure is available. Traditional wired charging has many problems in marine environments:

- Metal connectors rust and corrode in salt water
- Handling electrical cables near water creates safety risks
- Physical connections wear out with repeated use
- Docking must be very precise to connect cables
- Shore-side infrastructure is expensive to build

Wireless power transfer offers a solution to these problems because it removes the physical connection. But wireless charging for boats brings its own challenges. The boat moves with waves, which changes the distance and alignment between coils. The charging station must work even when sunlight is not available. And the whole system must be affordable to build and operate.

3.2 Specific Challenges Addressed

Challenge 1: Intermittent Solar Power

Solar panels only produce power when the sun is shining. Cloudy days and nighttime hours produce no solar energy. To solve this, our system includes a battery to store energy and a backup power demonstration using a simple fuel cell concept.

Challenge 2: Coil Distance and Alignment

The efficiency of wireless power transfer decreases as the distance between coils increases. It also decreases when coils are not properly aligned. Our experiments measure how efficiency changes with distance and misalignment to understand these effects.

Challenge 3: System Monitoring

Users need to know how much power is being transferred. Our system includes an ESP32 microcontroller with sensors and an LCD display to show voltage, current, and power in real time.

Challenge 4: Cost

Many existing WPT systems use expensive components. Our project uses affordable, commonly available parts to demonstrate that a practical system can be built at low cost.

3.3 Objectives

Objective 1: Design a simple wireless power transfer system using resonant inductive coupling that can send power across a small air gap.

Objective 2: Power the system using solar energy with an MPPT controller to extract maximum power from the solar panel.

Objective 3: Build a real-time monitoring system using an ESP32 microcontroller that measures voltage, current, and power and displays these values on an LCD screen.

Objective 4: Experimentally measure how power transfer efficiency changes with different coil distances and misalignment angles.

Objective 5: Demonstrate backup power options including onboard solar panels and a simple hydrogen fuel cell concept using magnesium and graphite electrodes.

Objective 6: Create a complete working prototype that shows the feasibility of solar-powered wireless charging for small electric boats.

Chapter 4: Methodology

4.1 Overall Approach

This project follows a systematic approach from component selection to final testing. The methodology is divided into several stages: design, component selection, assembly, programming, calibration, testing, and analysis.

4.2 System Working Principle

The complete system works in the following sequence:

Step 1: Solar Energy Capture

A solar panel converts sunlight into electrical energy. The panel is placed in direct sunlight during testing.

Step 2: Maximum Power Extraction

An MPPT charge controller adjusts the electrical load on the solar panel to keep it operating at its maximum power point. This extracts the most energy possible from the panel under current light conditions.

Step 3: Energy Storage

The electricity from the solar panel charges a 12-volt lead-acid battery. This battery stores energy for use when the sun is not shining or when a boat needs charging.

Step 4: Power Conversion for Wireless Transfer

The battery provides DC power to a high-frequency inverter. This inverter converts the DC power to high-frequency AC power. Higher frequencies allow more efficient wireless transfer with smaller coils.

Step 5: Magnetic Field Generation

The high-frequency AC current flows through a transmitter coil. This coil creates a changing magnetic field around it. The field extends outward from the coil.

Step 6: Magnetic Field Reception

A receiver coil placed near the transmitter coil captures some of this magnetic field. The changing magnetic field induces an AC voltage in the receiver coil.

Step 7: Power Conditioning

The induced AC voltage is converted to DC using a rectifier circuit. A voltage booster then raises the DC voltage to the level needed for charging.

Step 8: Real Time Monitoring

Voltage and current sensors measure the power at the receiver output. An ESP32 microcontroller reads these sensors and calculates power. The values are displayed on an LCD screen.

4.3 Operational Flowchart

The system operates in a continuous loop:

1. Start and initialize all components
2. Read solar panel voltage and current
3. Run MPPT algorithm to adjust operating point
4. Charge battery if power is available
5. Check if a boat is present for charging
6. If boat detected, activate transmitter coil
7. Measure receiver voltage and current
8. Calculate and display power on LCD
9. Repeat the cycle

4.4 Experimental Setup

Testing was conducted in the following manner:

1. **No-Load Test:** Measure receiver voltage with no load connected
2. **Load Test:** Connect resistive loads and measure power transfer
3. **Distance Test:** Vary the separation between coils from 1 cm to 6 cm
4. **Misalignment Test:** Shift coils sideways while keeping distance constant
5. **Solar Test:** Measure charging performance under sunlight
6. **Battery Test:** Verify battery charging and discharging operation

Chapter 5: System Design and Block Diagram

5.1 System Architecture

The complete system is divided into two main subsystems: the shore charging station and the boat receiver unit. Each subsystem contains several functional blocks.

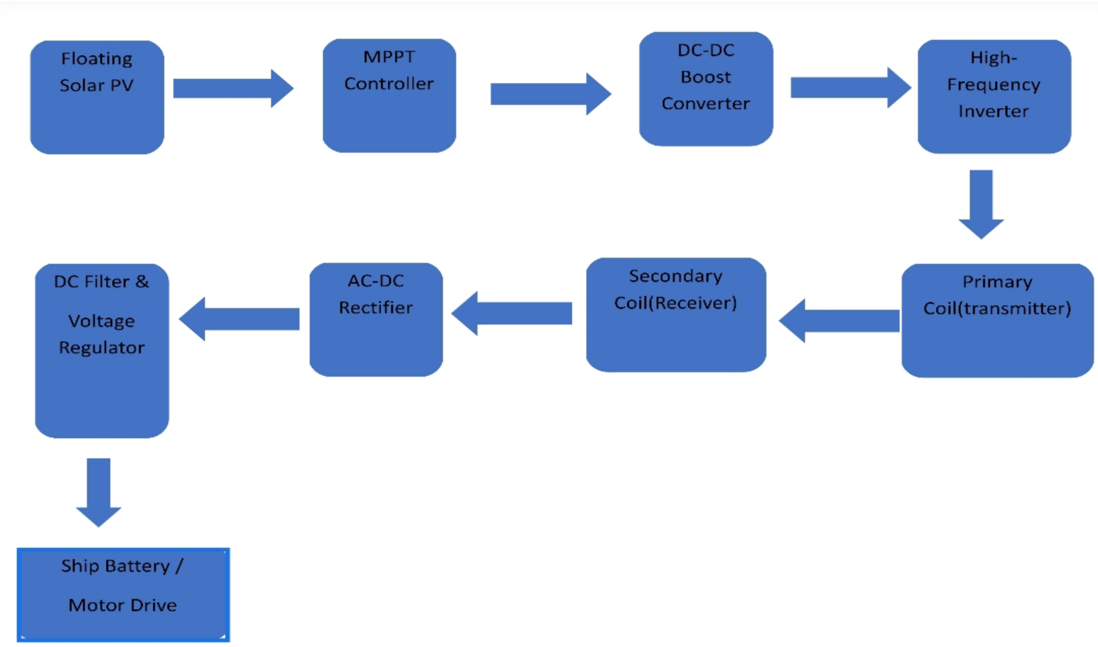

Fig.5.1.1 Block Diagram of System Architecture

5.2 Charging Station Subsystem

The charging station has five main components:

1. Solar Panel Array

Sunlight hits the panel and generates DC electricity. A 50-watt polycrystalline panel is chosen here. Polycrystalline cells cost less than monocrystalline ones and still perform decently under hazy or cloudy skies common near water bodies. The panel's open-circuit voltage is around 22V, which drops to about 18V under load.

2. MPPT Charge Controller

The MPPT controller does two jobs. First, it constantly adjusts the electrical load seen by the solar panel so that the panel operates at its maximum power point (the voltage where wattage peaks). Second, it prevents the battery from overcharging by switching to a lower charging current once the battery is full. A 20A rated controller is overkill for a 50W panel, but it leaves room for future expansion.

3. Battery Bank

A 12V, 7Ah lead-acid battery acts as an energy buffer. When the sun is strong, the battery charges. When a boat arrives for charging or when sunlight fades the battery supplies power to the transmitter. This decouples energy collection from energy delivery. A 7Ah capacity can run the transmitter at full power for roughly 1.5 hours.

4. High-Frequency Inverter

The battery outputs DC, but wireless power transfer works far better with high frequency AC (typically 50–200 kHz). A simple push-pull inverter built with MOSFETs converts the 12V DC into a square wave AC. The frequency is set by a pulse generator (a 555 timer or ESP32-generated PWM). Higher frequency allows smaller coils and capacitors for the same power level.

5. Transmitter Coil

The coil is made from enamelled copper wire (around 1mm thickness) wound into a flat spiral of 15 cm diameter and 10 turns. A 0.1µF polypropylene capacitor is connected in parallel with the coil, forming a resonant LC tank circuit. When the inverter drives this tank at its natural resonant frequency, the coil current and magnetic field become much stronger than in a non-resonant design.

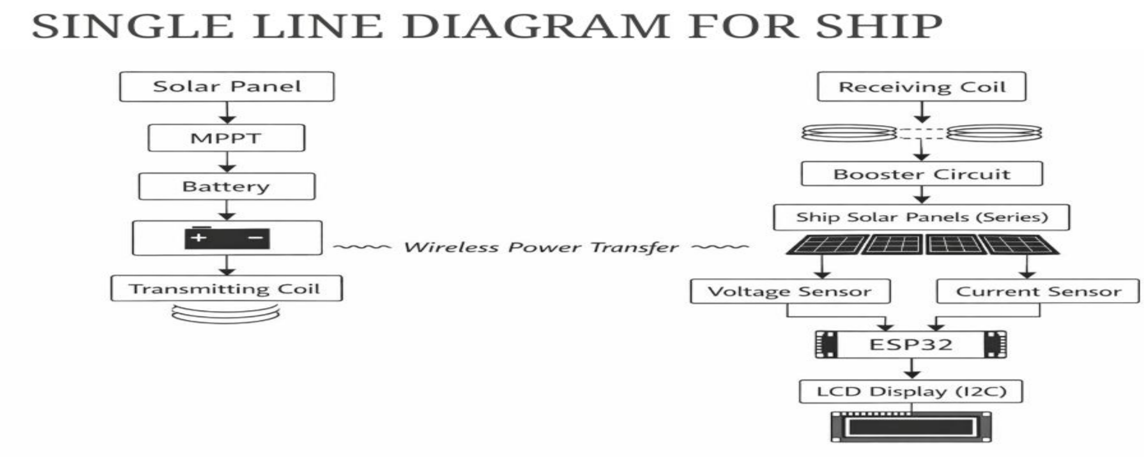

Fig.5.2.1 Single Line Diagram of Solar Wireless Power Transfer System for Electric Ship

5.3 Boat Receiver Subsystem

The boat receiver has four main components:

1. Transmitter Coil and Receiver Coil

The receiver coil is identical to the transmitter coil. It is mounted on the bottom of the boat model. When placed near the transmitter coil, it captures the magnetic field.

Fig5.3.1 Transmitter Coil and Receiver Coil

2. Voltage Booster

The rectified voltage may be lower than needed for battery charging. A DC-DC boost converter raises the voltage to the required level.

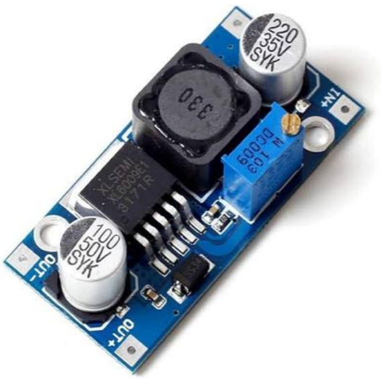

Fig5.3.2 Voltage Booster

4. Monitoring System

- **Voltage sensor**: A resistive divider (30kΩ and 7.5kΩ) scales the 0–15V range down to 0–3V for the ESP32's ADC.

Fig 5.3.3 Voltage Sensor

- **Current sensor**: ACS712 (20A version) outputs an analog voltage proportional to current. It measures current flowing into the boat's battery.

Fig 5.3.4 Current Sensor

- **Microcontroller**: ESP32-WROOM-32 reads both sensors every 500 ms, calculates power (voltage × current), and sends the data to the display.

Fig 5.3.5 ESP32

- **LCD display**: 16×2 character display with I2C backpack (only two wires to the ESP32). Shows real-time voltage, current, and power.

  The ESP32 can also be programmed to stop charging if the battery becomes full or if fault is detected.

Fig 5.3.6 LCD Display

5.5 Complete Component List:

Table 5.5 List of components

	Components	**Specification**	**Quantity**
1	Solar Panel	50W, 12V	4
2	MPPT Controller	20A, 12V/24V (PWM type also works but less efficient)	1
3	Battery	12V, 7Ah lead acid (sealed maintenance-free)	2
4	ESP32	ESP32-WROOM-32 (WiFi/Bluetooth optional for logging)	2
5	LCD Display	16×2, I2C interface (address 0x27 or 0x3F)	2
6	Current Sensor	ACS712 20A (outputs 100mV per ampere)	2
7	Voltage Sensor	30kΩ / 7.5kΩ divider (max 15V input)	2
8	Transmitter Coil	10 turns, 15cm diameter, 1mm copper wire	2
9	Receiver Coil	10 turns, 15cm diameter, 1mm copper wire	2

Chapter 6: Hardware Implementation and ESP32 Code

6.1 ESP32 Pin Connections

The ESP32 board needs to talk to two sensors and one LCD screen. Choosing the right pins matters because some pins have special functions. GPIO34 and GPIO35 are good choices for analog readings because they are pure input pins they cannot accidentally become outputs and cause short circuits. The I2C pins (GPIO21 and GPIO22) are standard on almost every ESP32 board, so library examples work without changes.

Table 6.1: ESP32 Pin Configuration

ESP32 Pin	**Connections**	**What it does**
GPIO34	Voltage sensor output	Reads the divided voltage (0 to 3.3V)
GPIO35	Current sensor output	Reads the ACS712 signal (around 2.5V at zero current)
GPIO21	LCD SDA	Carries I2C data to the display
GPIO22	LCD SCL	Carries I2C clock signal
3.3V	Sensor power	Supplies 3.3 volts to both sensors
GND	Common ground	Completes all circuits

6.2 Circuit Schematic

The monitoring part of the project breaks down into three simple blocks. Each block does one job, and together they give a complete picture of what is happening during charging.

Voltage Measurement Block

The boat battery can go up to 15 volts when fully charged. But the ESP32's analog pins only tolerate 0 to 3.3 volts. Feeding 15V directly would destroy the chip. A voltage divider solves this problem. Two resistors 30kΩ on top and 7.5kΩ on the bottom cut the voltage down by a factor of about 5. When the battery is at 15V, the divider outputs $15 \times (7.5 / (30 + 7.5)) = 3.0V$, which is safe. A small 10µF electrolytic capacitor across the output smooths any ripples.

Current Measurement Block

The ACS712 current sensor is a readymade module. It has three pins: VCC, GND, and OUT. The OUT pin gives 2.5V when no current flows. For every ampere of current in the positive direction, the voltage rises by 100mV. So at 2A, the output is 2.7V; at 5A, it would be 3.0V. The system rarely exceeds 2A, so we connect OUT directly to GPIO35 without extra resistors.

Display Interface Block

The 16×2 LCD with an I2C backpack is a huge time-saver. Without the backpack, you would need to wire 6 or 8 separate data lines plus control lines. With the backpack, only four wires are needed: VCC, GND, SDA, and SCL. The SDA line connects to GPIO21, and SCL connects to GPIO22.

6.3 Sensor Calibration

The ESP32's analog to digital converter (ADC) turns a voltage into a number between 0 and 4095. But that number by itself means nothing. Calibration finds the mathematical relationship between the raw number (or the measured voltage) and the real physical quantity either actual battery voltage or actual current.

1. Voltage Sensor Calibration

To find this relationship, we used a bench power supply that can output any voltage from 0 to 15V. We set it to eight different values, from 1V up to 15V. At each setting, we measured the voltage after the divider using a digital multimeter. The results are in Table 6.2.

Table 6.2: Voltage Sensor Calibration Data

Actual Voltage (V)	Sensor Output (V)
1.0	0.26
3.0	0.65
5.0	1.04
7.0	1.42
9.0	1.81
11.0	2.20
13.0	2.59
15.0	2.98

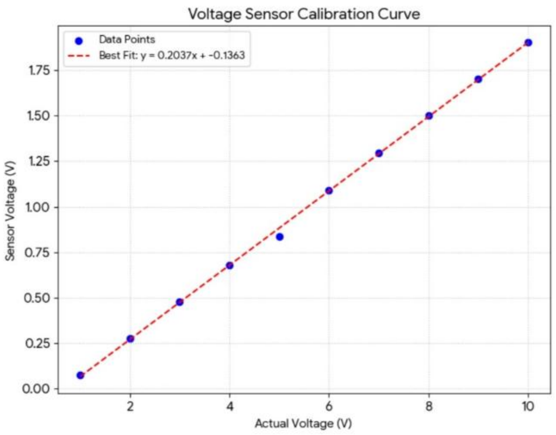

Fig 6.3.1 Voltage sensor calibration graph

Looking at the numbers, every time the actual voltage goes up by 2V, the sensor output goes up by roughly 0.39V. That is a consistent pattern. Plotting these points on graph paper gives a straight line. The line's equation comes out as:

Actual Voltage = (Sensor Voltage × 5.15) − 0.3

2. Current Sensor Calibration

The current sensor was calibrated by connecting a resistor load and a second multimeter in series. We varied the load and recorded both the reference current (from the multimeter) and the ACS712 output voltage. The test covered currents from 0 to 0.6A, which is the normal range for charging a small 7Ah battery.

Table 6.3: Current Sensor Calibration Data

Actual Current (A)	**Sensor Output (V)**
0.0	2.48
0.1	2.49
0.2	2.50
0.3	2.51
0.4	2.52
0.5	2.53
0.6	2.55

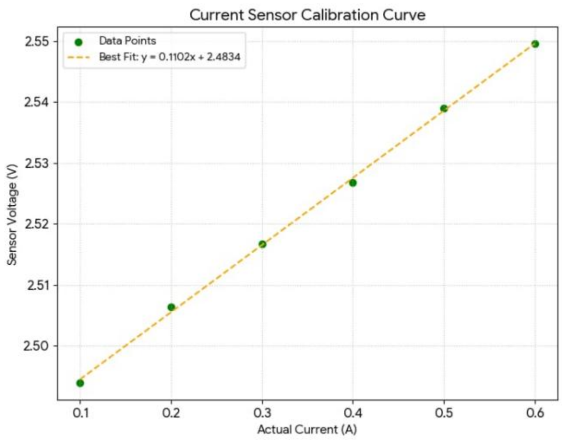

Fig 6.3.2 Current sensor Calibration Graph

Two things stand out. First, the zero-current reading is 2.48V, not exactly 2.5V as the datasheet promises. This is normal every chip has small offsets. Second, the voltage increases by about 0.11V for every 1A of current, giving a sensitivity of 0.1107 V/A. Using these two numbers, the formula becomes:

Current (A) = (Sensor Voltage − 2.484) / 0.1107

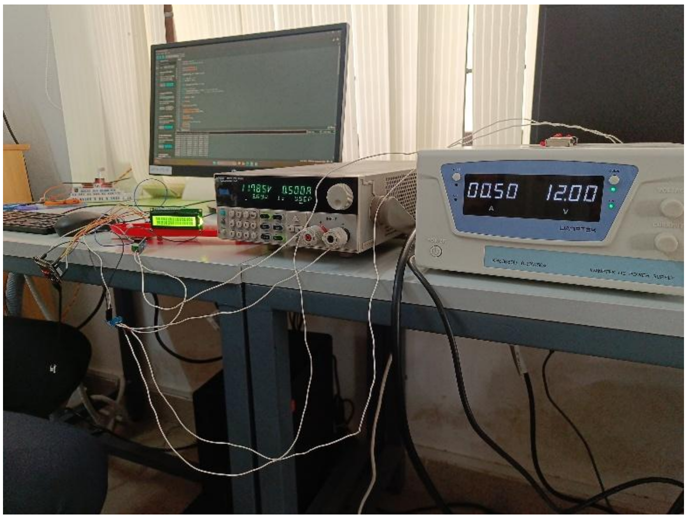

Fig 6.3.2 Voltage and current sensor calibration

6.4 ESP32 Code

The firmware for the ESP32 handles all measurement, calculation, and display tasks. While the overall structure is straightforward, several details are critical for obtaining stable and accurate readings. This subsection describes the program logic, explains key design choices, and presents the complete code.

1. Smart Program Algorithm and Working Process

The software follows a continuous loop structure. Upon power up or reset, the ESP32 performs a one-time setup phase. After setup, the main loop repeats indefinitely at a fixed rate.

Setup Phase:

The serial communication port is started at 115200 baud for debugging output.

The I2C LCD is initialized, its backlight is turned on, and a short startup message ("Boat Charging / Monitor Ready") is shown for two seconds.

The LCD screen is cleared to prepare for live data.

The analog to digital converter (ADC) is configured for 12-bit resolution (values from 0 to 4095) and an input range of 0 to 3.3 volts.

Main Loop:

The voltage sensor pin (GPIO34) is read 2000 times in quick succession. All readings are summed.

The current sensor pin (GPIO35) is also read 2000 times, and those readings are summed separately.

Each sum is divided by 2000 to obtain an average raw ADC value.

The average raw value is converted to a voltage (0–3.3V) using the formula: Voltage = (average Raw / 4095) × 3.3

The voltage sensor calibration equation is applied to obtain the actual battery voltage.

The current sensor calibration equation is applied to obtain the actual charging current.

Any negative current value (possible due to noise near zero) is set to zero to avoid misleading displays.

Power is calculated as: Power = Voltage × Current

The results are displayed on the LCD: first row shows voltage and current, second row shows power.

The same values are printed to the Serial Monitor for debugging and data logging.

The program pauses for 500 milliseconds, then the loop restarts.

2. Design Decisions and Justification

The wireless charging environment introduces several sources of electrical noise. The inverter operates at high frequency (tens of kHz), the boost converter switches rapidly, and the magnetic fields from the coils can induce small voltages in nearby wires. These noise sources cause the ADC readings to fluctuate from one measurement to the next. Taking 2000 samples and averaging them smooths out these random fluctuations. Testing showed that with 10 or 50 samples, the displayed values still jumped by ±0.2V and ±0.05A even under steady conditions. Increasing to 2000 samples reduced the jitter to less than ±0.02V and ±0.01A. The ESP32 can perform 2000 analog reads in approximately 20–30 milliseconds, which does not slow down the overall update noticeably.

The LCD can refresh much faster up to tens of times per second. However, a human observer cannot track numbers that change that rapidly. A 500 ms delay (two updates per second) gives the eye enough time to read each value comfortably. At the same time, the response feels immediate enough to observe changes when the charging current varies (for example, when a boat moves slightly off the charging pad). Slower updates (1 second or more) would miss brief fluctuations that might indicate alignment problems.

A hardware low pass filter (an RC circuit) could also reduce noise. However, software averaging offers two advantages. First, it requires no extra components only a few lines of code. Second, the number of samples can be changed easily during testing to find the optimum value. Hardware filters would require soldering different resistor or capacitor values.

The ESP32 firmware carries out the following sequence of operations. Some tasks happen only once at startup. Others repeat continuously in a loop.

Startup / Setup Phase (executed once):

1The LCD screen is powered on, the I2C connection is established, and a brief welcome message appears for two seconds.

2The analog-to-digital converter is configured for 12 bit resolution (0 to 4095 counts) and set to accept input voltages between 0 and 3.3 volts.

3.A WiFi access point is created with the network name "ElectricShip" and password "12345678". The ESP32 begins broadcasting its own wireless network.

4.A lightweight web server is started on port 80. The server listens for incoming browser requests.

5.The ESP32 prints its access point IP address (usually 192.168.4.1) to the serial monitor so the user knows where to point a web browser.

Main Loop:

6.The voltage sensor pin (GPIO34) is read 2000 times in a row. All 2000 values are added together.

7.The current sensor pin (GPIO35) is also read 2000 times, and those readings are summed separately.

8.Each sum is divided by 2000 to get an average raw ADC value for voltage and another for current.

9.The two averaged raw numbers are converted to actual voltages (0 3.3V range) using the formula:

pinVoltage = (averageRaw / 4095) × 3.3

10.The voltage calibration equation (from Section 6.3.1) turns the pin voltage into the real battery voltage.

11.The current calibration equation (from Section 6.3.2) turns the current sensor's pin voltage into the real charging current.

12.If the calculated current comes out negative (possible from noise when no current flows), it gets set to zero to avoid confusing the user.

13.Power is figured out by multiplying the true voltage by the true current: Power = Voltage × Current

14.The LCD screen updates. The top row shows voltage and current. The bottom row shows power.

15.The same three numbers (voltage, current, power) are printed to the serial monitor for debugging and record keeping.

16.The web server checks for any incoming browser requests. If a client (phone, laptop, tablet) asks for the web page, the ESP32 builds an HTML page containing the latest voltage, current, and power values and sends it back.

17.Any connected browser that loads the page will see a dashboard. A meta refresh tag inside the HTML makes the browser reload the page automatically every two seconds, giving live updates.

18.The whole cycle pauses for half a second (500 milliseconds), then jumps back to step 6 and repeats.

Chapter 7: Maximum Power Point Tracking

7.1 MPPT

A solar panel behaves differently depending on what it is connected to. Think of it like a water pump. If the pump outlet is completely blocked (open circuit), water pressure builds up but nothing flows. No work gets done. If the pump outlet is wide open with no hose (short circuit), water flows fast but there is no pressure. Again, no useful work.

The same idea applies to a solar panel. At one extreme, with nothing attached, the panel shows high voltage but zero current. Power (voltage × current) equals zero. At the other extreme, with the wires touched together, current flows but voltage drops to nearly zero. Power is zero again.

Somewhere between these two extremes lies a sweet spot. At that voltage, the panel delivers the most possible power. Engineers call this the Maximum Power Point (MPP). Every solar panel has one, and its location changes with conditions.

When we take a 50W panel like the one used in this project. On a sunny afternoon, the open-circuit voltage might be 22V. The short circuit current might be 3A. If a variable resistor I connected and adjusted slowly, there will be a power start low, climb to a peak, then fall again. The peak might happen at roughly 17V and 2.9A, giving about 49W. That 17V is the maximum power point for those conditions.

7.2 Why MPPT is Needed

The maximum power point does not stay in one place. Two main factors cause it to move around.

Sunlight intensity changes

On a bright sunny day, the panel produces lots of current. The maximum power point sits at a higher voltage and higher current compared to a cloudy day. In the morning and evening, when sunlight slants through more atmosphere, the MPP shifts lower. Without a tracking system, the panel would often operate at the wrong voltage, wasting a significant portion of its potential output.

Temperature effects

Solar panels actually work worse when they get hot. A panel lying in the summer sun can reach 60°C or more. As temperature rises, the maximum power point voltage drops. On a cool winter morning, the MPP voltage might be 18V. On a hot summer afternoon, the same panel might have its MPP at only 15V. The current changes only slightly with temperature, so the voltage shift is the main concern.

If you connect a solar panel directly to a battery, the battery forces the panel to run at the battery's voltage. A 12V lead acid battery might sit at 13.8V when charging. If the panel's MPP that moment is actually 17V, the panel gets dragged away from its best operating point. The result is lost power. Tests show that a direct connection can waste 20-30% of the energy the panel could otherwise deliver. An MPPT controller recovers most of that lost energy.

In this boat charging project, a 30% improvement means the 50W panel acts more like a 65W panel (without needing a larger, more expensive panel). Over a full day of sunlight, the extra energy can fill the station battery faster or charge an extra boat.

7.3 Perturb and Observe Algorithm

There are several ways to track the maximum power point. This project uses a method called Perturb and Observe (P&O). It is popular because it works reliably and does not require complicated math or expensive sensors.

Step by step walkthrough

Imagine you are trying to find the best volume setting on a radio, but you cannot see the numbers. You can only listen and tell if it gets louder or quieter. The P&O algorithm works the same way:

1. Measure the panel's current voltage and current. Multiply them to find the current power.
2. Make a tiny adjustment to the operating voltage. On a radio, this is like turning the volume knob a little bit one way.
3. Measure again and calculate the new power.
4. Compare the new power to the old power.
   - If the new power is higher than the old power, you turned the right way. Keep turning in the same direction.
   - If the new power is lower, you turned the wrong way. Reverse direction for the next adjustment.
5. Repeat this process continuously, even after finding the peak. The peak can drift over time, so the algorithm never stops checking.

As long as the sunlight changes slowly (seconds or minutes), the P&O algorithm keeps the panel near its true maximum power point. The small perturbations cause the operating point to dance around the peak slightly, but the average power stays high.

Limitations of P&O

The algorithm can get confused if sunlight changes very quickly. For example, a cloud suddenly covering the panel drops the power for reasons unrelated to the perturbation. The algorithm might then move the operating point in the wrong direction. However, for most real-world conditions with typical cloud movements, P&O performs well enough.

Flowchart representation

The decision path the controller follows. It starts with measuring, then calculates, then decides whether to increase or decrease the voltage, then loops back to the beginning.

7.4 MPPT Implementation in This Project

Instead of building an MPPT circuit from scratch, this project uses a ready made commercial controller.

This approach saves time and ensures reliable operation.

The chosen controller has a rated current of 20A and accepts 12V or 24V battery banks. It connects between the solar panel and the station battery. Inside the controller, a microprocessor runs the P&O algorithm described above. The controller measures panel voltage and current dozens of times per second, adjusts its internal switching regulator, and keeps the panel at its maximum power point.

Additional protection features

Besides tracking the MPP, the controller also acts as a battery charger. It monitors the station battery voltage. When the battery becomes full (typically 14.4V for a 12V lead-acid battery), the controller reduces the charging current to a safe level. This prevents overcharging, which can boil away the electrolyte and ruin the battery. Once the battery voltage drops a little (for example, after powering the inverter for a while), the controller resumes full charging.

The wiring is straightforward:

- Solar panel positive and negative wires go to the controller's "PV input" terminals.
- Station battery positive and negative go to the controller's "battery" terminals.

- The load terminals (sometimes labeled "LOAD") are not used in this project because the inverter connects directly to the battery.

The controller has a small LCD screen on its front panel. This screen shows the panel voltage,
battery voltage, charging current, and a small graphic indicating the MPPT status. For this project, the user does not need to interact with the controller once it is set up. It runs automatically.

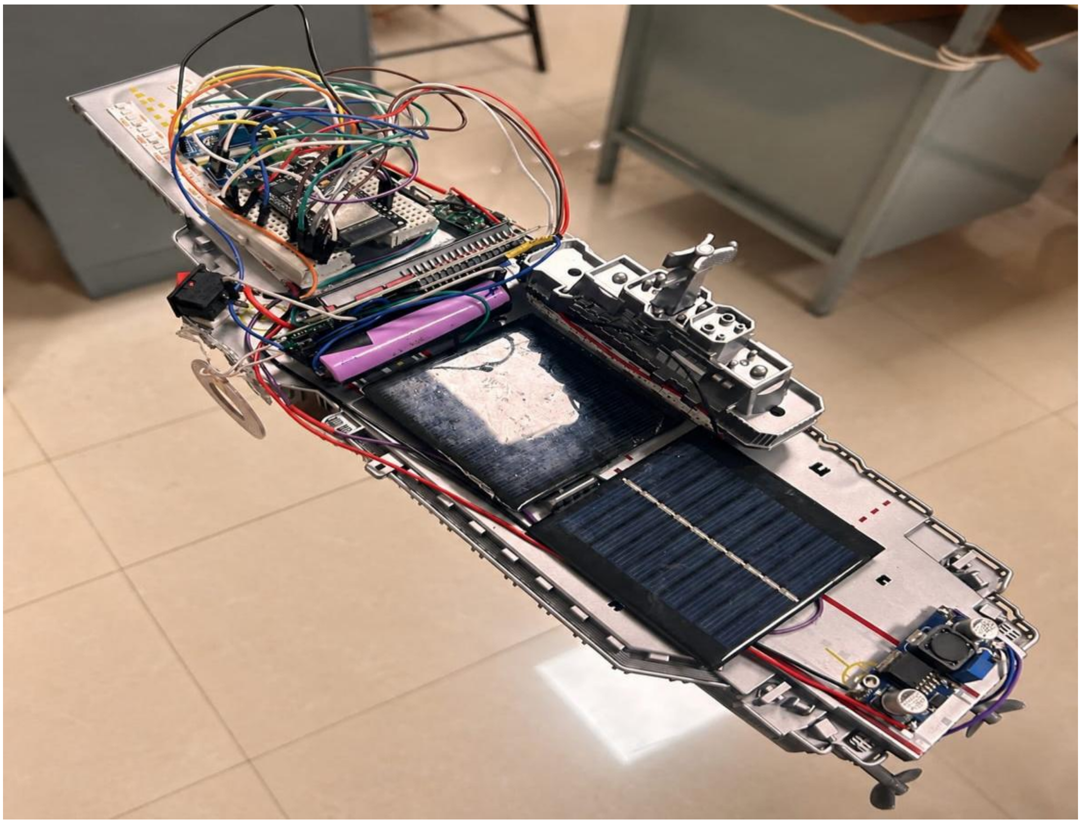

Fig 7.4.1 Solar Powered Smart Ship Prototype

Chapter 8: Innovative Functions of the Charging System

This chapter describes several additional features built into or demonstrated with the wireless boat charging system. Some functions are fully implemented, while others are shown as proof-of-concept demonstrations.

8.1 Battery Status Monitoring

The charging station keeps track of the storage battery's voltage at all times. Voltage level is a good indicator of how much energy remains in the battery. The ESP32 reads the battery voltage through the voltage divider circuit described in Chapter 6, then shows the value on the LCD.

Two common problems can damage lead-acid batteries if left unchecked. The first is overcharging. When a battery reaches full charge (around 14.4V for a 12V lead-acid battery), continuing to push current into it causes the electrolyte to boil away and the internal plates to warp. The MPPT charge controller prevents this by automatically reducing or stopping the charging current once the voltage hits a safe upper limit.
The second problem is deep discharge. Draining a lead acid battery below about 10.5V causes permanent damage. The system does not automatically disconnect the load in this version, but the LCD warns the user when voltage drops into the danger zone. A future improvement could add a relay that cuts off the transmitter coil when the battery gets too low.

The LCD screen updates twice per second. A typical reading might show "V=12.8V" on the top row. An experienced user knows that 12.6V to 12.8V means roughly 80-100% charge, while 11.8V to 12.0V indicates the battery is nearly empty.

8.2 Wireless Charging from Station to Boat

This is the core function of the whole project. Instead of metal contacts or cables, power travels across a short air gap using magnetic fields.

The charging station inverter sends high frequency AC (around 100 kHz) through the transmitter coil. This coil sits flush with the dock surface. The boat has an identical coil mounted on its underside. When the boat is parked directly above the transmitter coil, the magnetic field induces a voltage in the receiver coil. That voltage gets rectified, boosted, and fed into the boat's own battery.

Traditional charging methods require plugging and unplugging connectors. Around water, electrical connectors can corrode quickly. They also pose a shock risk if water gets into the plug. Wireless charging has no exposed conductors. The transmitter coil can even be sealed inside a waterproof plastic enclosure. The boat's receiver coil can be encapsulated in epoxy. This makes the whole system much safer for marine environments.

Measured performance

In the laboratory prototype, with the coils spaced 10 mm apart and centered properly, the system delivered approximately 8-10 watts to the boat battery. Efficiency measured from the station battery to the boat battery was around 65-70%. The remaining energy turned into heat in the coils and electronics.

8.3 Onboard Solar Panels

The charging station is not the only place where solar energy can be harvested. Small flexible solar panels can be attached to the boat's deck or cabin roof.

How onboard panels help

While the boat is moving during daylight hours, the onboard panels generate electricity directly into the boat's battery. This extends the time between visits to the charging station. For example, a boat that normally needs charging every two hours might run for three or four hours with an onboard 10 watt panel helping out.

Practical limitations

On a small boat model, the available surface area limits the panel size. A typical model boat might fit a 5W or 10W panel. This is not enough to run a motor continuously, but it can top up the battery during stops or slow cruising. Larger real world boats could mount several hundred watts of panels on cabin roofs.

Integration with the system

The onboard panels connect to the boat's battery through a small charge controller (a simple PWM type is sufficient for small panels). The ESP32 monitoring system already measures the boat battery voltage and charging current, so it automatically shows the combined effect of wireless charging plus solar charging.

8.4 Hydrogen Fuel Cell Backup

A small demonstration was built to show how a hydrogen fuel cell could serve as a backup power source. This is not a full size fuel cell, but rather a simple classroom style experiment that illustrates the principle.

Demonstration setup

Two different metal electrodes are placed in a glass of salt water (sodium chloride solution). One electrode is made of magnesium ribbon. The other is a graphite rod (like the kind found in a standard pencil). A voltmeter connected across the two electrodes shows a reading of roughly 0.8 to 1.2 volts.

The chemical reaction

Magnesium is more reactive than carbon. In salt water, the magnesium slowly dissolves, releasing electrons. These electrons travel through the external wire to the graphite electrode, where they react with water and dissolved oxygen. The salt in the water acts as an electrolyte, allowing ions to move between the two electrodes. This creates a small but measurable electric current.

What this demonstrates

The demonstration proves that electricity can be generated from chemical reactions without combustion. A real hydrogen fuel cell works on a similar principle but uses hydrogen gas and oxygen as the reactants, with a special membrane separating the electrodes. Real fuel cells produce much higher voltage (typically 0.6-0.8V per cell) and can be stacked to reach 12V or more.

Potential for the boat system

If a true hydrogen fuel cell were added to the charging station, it could provide backup power during extended cloudy periods. The fuel cell would run on stored hydrogen (in a tank) and produce only water vapor as exhaust. This keeps the system environmentally clean. However, hydrogen fuel cells remain expensive, so this feature is suggested only as a future upgrade.

8.5 Nighttime Charging Capability

Solar panels stop producing electricity when the sun goes down. A system without energy storage would be useless at night. The charging station solves this problem with its 12V, 7Ah lead-acid battery.

During sunny hours, the solar panel generates up to 50 watts. Part of that power may go directly to charging a boat if one is present. The rest goes into the station battery. The MPPT charge controller manages this split automatically. By late afternoon, the station battery is typically full.

Nighttime operation

When a boat arrives for charging after dark, the station battery supplies all the needed power. The inverter draws from the battery, not from the solar panel (which is idle). A full 7Ah battery at 12V stores about 84 watt hours. The system draws roughly 10-12 watts while charging a boat. Simple math shows about 7-8 hours of continuous charging possible from a fully charged station battery.

Limitations

If multiple boats need charging overnight, or if a single boat requires a very long charge, the station battery may run empty. The LCD shows the station battery voltage dropping over time. The user can decide whether to continue or wait for daytime solar recharging. For practical use, a larger station battery (say 20Ah or 40Ah) would provide more nighttime reserve.

24 hour capability statement

As long as the station battery holds charge, the system can deliver wireless charging at any hour. This makes it far more useful than a direct solar-only system that works only when the sun shines.

Chapter 9: Results and Discussion

This chapter presents the actual measurements taken from the working prototype. All tests were done indoors using a 12V power supply and also outdoors using the solar panel, depending on the specific experiment.

9.1 Experimental Setup

All experiments were carried out on a laboratory bench before any outdoor testing. The setup consisted of the following items placed on a wooden table:

- The charging station with its 12V, 7Ah battery (or a DC power supply set to 12V for repeatable tests)
- The high-frequency inverter board connected to the transmitter coil
- The receiver coil mounted on a small wooden frame that could be raised and lowered to change the air gap
- A resistive load (set of power resistors) or a small 12V battery as the boat load
- The ESP32 monitoring system connected to voltage and current sensors
- A digital multimeter used as a reference to check the ESP32 readings

Coil positioning method

The transmitter coil was fixed to the table surface. The receiver coil was attached to a sliding stand made from a threaded rod and a wing nut. Turning the nut raised or lowered the receiver coil in 1mm steps. A ruler was taped to the stand to measure the gap distance between the two coils.

Load conditions

Two types of loads were used during testing. For efficiency measurements, a purely resistive load (a 10Ω, 20W resistor) was connected to the output of the receiver's voltage booster. This gave a steady, predictable current draw. For realistic testing, a small 12V, 2.2Ah sealed lead-acid battery was connected instead. The battery started at roughly 50% charge for each test.

Data recording method

The ESP32 printed voltage, current, and power to the Serial Monitor every 0.5 seconds.

While running each test, the Serial Monitor output was copied into a text file. Later, the data was moved into a spreadsheet program to create tables and graphs. For each test condition, readings were recorded for 60 seconds, and the average values were calculated.

Variables tested

The following conditions were changed one at a time:

1. Air gap distance between coils (from 0 mm contact up to 30 mm)
2. Horizontal misalignment (receiver coil shifted sideways by 0, 10, 20, 30 mm)
3. Load current (changed by swapping resistors or using different battery states)
4. Input voltage to the inverter (10V, 11V, 12V, 13V, 14V)

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Appendix:

```
#include <WiFi.h>
#include <WebServer.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

        const int VOLTAGE_PIN = 34;    // Voltage sensor (after divider)
const int CURRENT_PIN = 35;    // ACS712 current sensor output

LiquidCrystal_I2C lcd(0x27, 16, 2);  // Address 0x27 (try 0x3F if not working)

const char* ssid = "ElectricShip";
const char* password = "12345678";
WebServer server(80);

 Voltage sensor: Actual Voltage = (Sensor Voltage × 5.15) - 0.33
const float VOLTAGE_SLOPE = 5.15;
const float VOLTAGE_INTERCEPT = -0.33;

const float CURRENT_ZERO_VOLTAGE = 2.484;
const float CURRENT_SENSITIVITY = 0.1107;

const float ADC_REFERENCE_VOLTAGE = 3.3;
const int ADC_MAX_CODE = 4095;      // 12-bit resolution
const int NUM_SAMPLES = 2000;       // Number of readings to average

float voltage = 0.0;
float current = 0.0;
float power = 0.0;

float readAverageVoltage(int pin);
void updateLCD();
void handleWebRoot();
void updateMeasurements();
```

```
void handleWebRoot() {
  String html = "<!DOCTYPE html><html>";
  html += "<head>";
  html += "<meta http-equiv='refresh' content='2'>";
  html += "<meta name='viewport' content='width=device-width, initial-scale=1'>";
  html += "<title>Electric Ship Monitor</title>";
  html += "<style>";
  html += "body { font-family: Arial, sans-serif; text-align: center; margin-top: 50px; }";
  html += "h1 { color: #1E88E5; }";
  html += ".value { font-size: 2em; margin: 20px; }";
  html += ".voltage { color: #2E7D32; }";
  html += ".current { color: #D32F2F; }";
  html += ".power { color: #FF8F00; }";
  html += "</style>";
  html += "</head><body>";
  html += "<h1>Electric Ship Charging Dashboard </h1>";
  html += "<div class='value voltage'> Voltage: " + String(voltage, 2) + " V</div>";
  html += "<div class='value current'>Current: " + String(current, 2) + " A</div>";
  html += "<div class='value power'>Power: " + String(power, 2) + " W</div>";
  html += "<p><small>Page auto-refreshes every 2 seconds</small></p>";
  html += "</body></html>";

  server.send(200, "text/html", html);
}

  void updateMeasurements() {

  float sensorVoltage = readAverageVoltage(VOLTAGE_PIN);
  voltage = (sensorVoltage * VOLTAGE_SLOPE) + VOLTAGE_INTERCEPT;

  float currentSensorVoltage = readAverageVoltage(CURRENT_PIN);
  current = (currentSensorVoltage - CURRENT_ZERO_VOLTAGE) /
CURRENT_SENSITIVITY;
```

```
  if (current < 0) {
    current = 0;
  }


  power = voltage * current;
}

float readAverageVoltage(int pin) {
  long total = 0;
  for (int i = 0; i < NUM_SAMPLES; i++) {
    total += analogRead(pin);
  }
  float averageRaw = (float)total / NUM_SAMPLES;
  float averageVoltage = (averageRaw / ADC_MAX_CODE) * 
ADC_REFERENCE_VOLTAGE;
  return averageVoltage;
}

void updateLCD() {
  lcd.setCursor(0, 0);
  lcd.print("V=");
  lcd.print(voltage, 2);
  lcd.print("V  I=");
  lcd.print(current, 2);
  lcd.print("A ");

  lcd.setCursor(0, 1);
  lcd.print("P=");
  lcd.print(power, 2);
  lcd.print("W     ");
}
```

```
void setup() {
  Serial.begin(115200);
  Serial.println("\nESP32 Boat Charging Monitor Starting...");


  Wire.begin();  // Default pins: SDA=21, SCL=22
  lcd.init();
  lcd.backlight();
  lcd.setCursor(0, 0);
  lcd.print("Boat Charging");
  lcd.setCursor(0, 1);
  lcd.print("Monitor Ready");
  delay(2000);
  lcd.clear();


  analogReadResolution(12);          // 12-bit (0-4095)
  analogSetAttenuation(ADC_11db);    // 0-3.3V input range


  WiFi.softAP(ssid, password);
  Serial.print("WiFi Access Point Started. IP Address: ");
  Serial.println(WiFi.softAPIP());


  server.on("/", handleWebRoot);
  server.begin();
  Serial.println

  delay(1000);
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("WiFi Ready");
  lcd.setCursor(0, 1);
```

```
  lcd.print("IP:192.168.4.1");
  delay(3000);
  lcd.clear();
}

  // Handle web client requests
  server.handleClient();

  // Update measurements
  updateMeasurements();

  // Update LCD display
  updateLCD();

  // Send data to serial monitor for debugging
  Serial.print("V: ");
  Serial.print(voltage, 2);
  Serial.print(" V | I: ");
  Serial.print(current, 3);
  Serial.print(" A | P: ");
  Serial.print(power, 2);
  Serial.println(" W");

  // Update every 500 milliseconds
  delay(500);
}
```

Appendix-A

DrillBit

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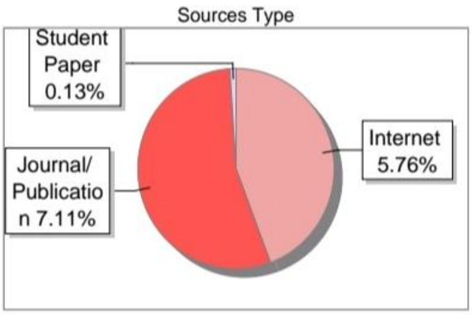

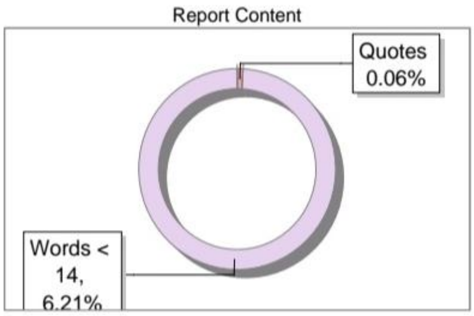

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Appendix- B

DrillBit

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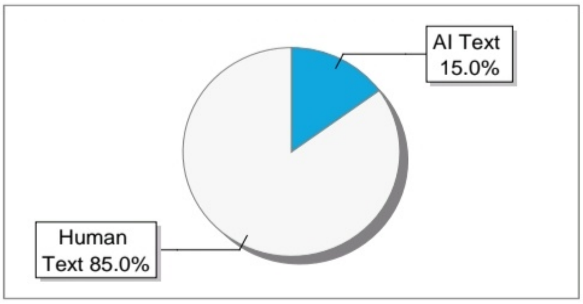

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Publication Details

Submission Summary

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Paper Title
Solar-Powered Wireless Power Transfer System for Electric Ships

Abstract
The Solar Wireless Power Transfer System for Ship is designed to provide an efficient and eco-friendly method for charging electric ships using renewable energy. In this system, solar panels installed at the charging station generate electrical energy. To ensure that the maximum possible power is extracted from the panels, an MPPT (Maximum Power Point Tracking) controller is employed the energy is then stored in a battery for later use. An ESP32 microcontroller is used as the main control unit of sysytem.it continuously monitors the operations using connected voltage and current sensors. These sensors measure the electrical parameters during power transfer, and the values are displayed on a 16×2 LCD for a real time observation. The efficiency of power transfer mainly depends on factors such as the distance between the coil, their alignment, and the angle at which the ship is positioned relative to the charging station and to any variation in these factors can affect the amount of energy received The ESP32 is programmed using the Arduino IDE, which manages sensors readings and display system. In addition to wireless charging, solar panels are also installed on the ship itself. These panels help us to back up the power source, helping to charge the battery when wireless charging is not available or when the battery level is too low.

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SDG

Mapping the Project to Sustainable Development Goals (SDGs) Sustainable Development Goals (SDGs)

The Sustainable Development Goals (SDGs) are a set of 17 global goals established by the United Nations to address the world's most pressing challenges, including poverty, inequality, climate change, environmental degradation, peace, and justice. These goals provide a framework for governments, institutions, and individuals to work together towards a sustainable and inclusive future for all. Engineering projects and technological innovations play a crucial role in achieving these goals by offering practical, scalable, and impactful solutions to real-world problems.

List of Sustainable Development Goals and Brief Description

SDG No.	Sustainable Development Goal	Brief Description
1	No Poverty	End poverty in all its forms everywhere
2	Zero Hunger	End hunger, achieve food security and improved nutrition, and promote sustainable agriculture
3	Good Health and Well-being	Ensure healthy lives and promote well-being for all at all ages
4	Quality Education	Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all
5	Gender Equality	Achieve gender equality and empower all women and girls
6	Clean Water and Sanitation	Ensure availability and sustainable management of water and sanitation for all

7	Affordable and Clean Energy	Ensure access to affordable, reliable, sustainable, and modern energy for all
8	Decent Work and Economic Growth	Promote sustained, inclusive, and sustainable economic growth and decent work for all
9	Industry, Innovation and Infrastructure	Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation
10	Reduced Inequalities	Reduce inequality within and among countries
11	Sustainable Cities and Communities	Make cities and human settlements inclusive, safe, resilient, and sustainable
12	Responsible Consumption and Production	Ensure sustainable consumption and production patterns
13	Climate Action	Take urgent action to combat climate change and its impacts
14	Life Below Water	Conserve and sustainably use oceans, seas, and marine resources
15	Life on Land	Protect, restore, and promote sustainable use of terrestrial ecosystems
16	Peace, Justice and Strong Institutions	Promote peaceful and inclusive societies, provide access to justice, and build effective institutions
17	Partnerships for the Goals	Strengthen global partnerships for sustainable development

Project–SDG Mapping

In alignment with the Vision and Mission of the Department to develop technically competent, socially responsible, and innovative engineer's students are required to map their project components to relevant Sustainable Development Goals (SDGs).
Each project should clearly demonstrate how its design, functionality, or implementation contributes to one or more SDGs.

Project–SDG Mapping Format

Project Title: Solar Powered Wireless Power Transfer System for Electric ship

SDG(s):

- SDG 7: Affordable and Clean Energy
- SDG 9: Industry, Innovation and Infrastructure
- SDG 11: Sustainable Cities and Communities
- SDG 13: Climate Action
- SDG 14: Life Below Water

Problem Solved:

Electric boats and ships currently rely on wired charging using cables and plugs. In marine environments, water and salt cause rapid corrosion of metal connectors, leading to equipment failure, high maintenance costs, and serious risk of electric shock. Additionally, most charging stations use electricity from fossil fuel power plants, which pollutes the air and water. There is no simple, safe, and clean way to charge electric boats without using physical cables.
This project solves this problem by creating a wireless charging system that uses solar energy. The boat does not need to plug into any cable. The system uses magnetic fields to transfer power from a charging station to the boat. The energy comes from the sun, which is free and clean. This makes boat charging safe, easy, and environmentally friendly.

Reason for Selecting the SDG(s):

- SDG 7 (Affordable and Clean Energy):
  The project uses solar panels to generate electricity, which is a clean and renewable energy source. It does not burn any fossil fuels.
- SDG 9 (Industry, Innovation and Infrastructure):
  The project introduces wireless power transfer technology to the marine industry. This is an innovative way to charge electric boats without physical infrastructure like cables and plugs.
- SDG 11 (Sustainable Cities and Communities):
  Electric boats charged by solar power do not produce smoke or noise. This makes coastal cities and communities cleaner, quieter, and healthier places to live.
- SDG 13 (Climate Action):
  By replacing diesel-powered boats with solar-charged electric boats, the project helps reduce carbon dioxide and other greenhouse gas emissions that cause climate change.
- SDG 14 (Life Below Water):
  Diesel boats can leak fuel and oil into the water, harming fish and other marine life. Solar-powered electric

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